MAC F244 Stochastic Calculus and Applications in Finance · Semester I, AY 2026–27

Continuous-Time Portfolio Optimization: The Merton Problem and the Cost of Estimation Risk

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1. Problem description

An investor divides wealth between a risky stock and a risk-free bank account and can rebalance continuously. *What fraction of wealth should be held in the stock to maximise the expected utility of final wealth?* This is the Merton problem (suggested topic 14). Under geometric Brownian motion and power utility, its answer is the constant fraction $\pi^* = (\mu - r)/(\gamma\sigma^2)$.

The formula requires the *true* drift μ and volatility σ, which are never known; an investor must estimate them from historical data. Our project therefore asks:

> *Does Merton's optimal strategy still perform well when its parameters are estimated from real market data, and how much expected utility does the estimation error cost?*

We propose to (i) derive the optimal strategy from the Hamilton–Jacobi–Bellman (HJB) equation, (ii) verify it numerically with SDE simulation, (iii) quantify the loss caused by parameter estimation in a controlled simulated market, and (iv) test the estimated strategy out of sample on Indian market data (NIFTY 50 and selected NSE stocks).

2. Mathematical model

Market. The stock follows geometric Brownian motion and the bank account grows at rate r:

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t, \qquad dB_t = rB_t\,dt,$$

where W_t is a standard Brownian motion. By Itô's lemma, $S_t = S_0 \exp\big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\big)$.

Wealth. If π_t is the fraction of wealth X_t held in the stock, self-financing gives

$$dX_t = X_t\big[r + \pi_t(\mu - r)\big]dt + X_t\,\pi_t\,\sigma\,dW_t.$$

Objective. With constant relative risk aversion (CRRA) utility $U(x) = x^{1-\gamma}/(1-\gamma)$, $\gamma > 0$, $\gamma \neq 1$, the investor solves $V(t,x) = \sup_\pi \mathbb{E}\big[U(X_T) \mid X_t = x\big]$.

HJB equation and solution. Dynamic programming and Itô's lemma give

$$0 = V_t + \sup_\pi \Big\{\big[r + \pi(\mu - r)\big]xV_x + \tfrac{1}{2}\pi^2\sigma^2x^2V_{xx}\Big\}, \qquad V(T,x) = U(x).$$

The first-order condition is $\pi^* = -(\mu - r)V_x/(\sigma^2 xV_{xx})$. The ansatz $V(t,x) = f(t)\,x^{1-\gamma}/(1-\gamma)$ gives $xV_{xx}/V_x = -\gamma$, hence

$$\boxed{\pi^* = \frac{\mu - r}{\gamma\sigma^2}} \qquad f(t) = \exp\Big((1-\gamma)\Big(r + \frac{(\mu-r)^2}{2\gamma\sigma^2}\Big)(T-t)\Big).$$

Certainty equivalent and cost of a wrong fraction. For any constant π, X_T is lognormal and $\mathbb{E}[U(X_T)] = U\big(x_0 e^{\mathrm{CE}(\pi)T}\big)$, where

$$\mathrm{CE}(\pi) = r + \pi(\mu - r) - \tfrac{1}{2}\gamma\sigma^2\pi^2$$

is the certainty-equivalent rate: the guaranteed continuously compounded annual rate the investor values equally to the risky strategy. It is a parabola maximised at π^*, so using an estimate $\hat{\pi}$ costs

$$\mathrm{Loss}(\hat{\pi}) = \mathrm{CE}(\pi^*) - \mathrm{CE}(\hat{\pi}) = \tfrac{1}{2}\gamma\sigma^2(\hat{\pi} - \pi^*)^2 \text{ per year},$$

compared with the benefit of optimal investing $\mathrm{CE}(\pi^*) - r = (\mu - r)^2/(2\gamma\sigma^2)$. Since $\mathrm{CE}(\hat{\pi}) - r = \tfrac{1}{2}\gamma\sigma^2\,\hat{\pi}\,(2\pi^* - \hat{\pi})$, the estimated strategy has a lower certainty-equivalent rate than holding cash exactly when

$$\hat{\pi}\,(2\pi^* - \hat{\pi}) < 0.$$

When $\mu > r$, this reduces to $\hat{\pi} < 0$ or $\hat{\pi} > 2\pi^*$.

Estimation. Daily log returns $R_i = \ln(S_i/S_{i-1})$ are i.i.d. $N\big((\mu - \tfrac{1}{2}\sigma^2)\Delta t,\ \sigma^2\Delta t\big)$ with $\Delta t = 1/252$. From the sample mean m and variance s^2, $\hat{\sigma}^2 = s^2/\Delta t$, $\hat{\mu} = m/\Delta t + \hat{\sigma}^2/2$ and $\hat{\pi} = (\hat{\mu} - r)/(\gamma\hat{\sigma}^2)$. Under constant-parameter GBM, L years of observations give approximately

$$\mathrm{se}(\hat{\mu}) \approx \frac{\sigma}{\sqrt{L}}$$

(only more *years* help, not more frequent sampling). Neglecting volatility estimation error gives the benchmark approximations

$$\mathrm{sd}(\hat{\pi}) \approx \frac{1}{\gamma\sigma\sqrt{L}}, \qquad \mathbb{E}[\mathrm{Loss}] \approx \frac{1}{2\gamma L},$$

where Loss is the certainty-equivalent rate loss defined above. The numerical experiments will also account for volatility estimation error.

3. Proposed computational methodology

All code will be written in Python (`numpy`, `pandas`, `scipy`, `matplotlib`, `yfinance`) as a shared, commented repository with a fixed random seed for reproducibility.

1. **Data and parameter estimation.** Daily prices from January 2008 for NIFTY 50 (`^NSEI`) and three NSE stocks from different sectors (HDFC Bank, Infosys, Hindustan Unilever) via Yahoo Finance; 91-day T-bill yields from the RBI Database on Indian Economy as the risk-free rate. Estimate $\hat{\mu}, \hat{\sigma}, \hat{\pi}$ with standard errors and 5-year rolling windows, and test the normality assumption (QQ plot, Jarque–Bera test).
2. **SDE simulation and verification.** Implement Euler–Maruyama and Milstein schemes for the price and wealth SDEs. Check strong and weak convergence against the exact GBM solution on a common Brownian path. Verify numerically, with common random numbers, that simulated expected utility over a grid of π peaks near π^* and agrees with the theoretical value $U\big(x_0 e^{\mathrm{CE}(\pi)T}\big)$ within Monte Carlo and discretisation error; study the sensitivity of π^* to γ, μ and σ.
3. **Estimation-risk experiment.** Fix "true" parameters equal to the full-sample estimates, so the true π^* is known. Generate 1,000 synthetic price histories of length $L \in \{5, 10, 20\}$ years, re-estimate $\hat{\pi}$ from each, and measure the distribution of $\hat{\pi}$, the average certainty-equivalent rate loss and the fraction of histories in which the estimated strategy is worse than cash. Compare with the theoretical predictions above. Repeat the experiments for $\gamma \in \{2, 3, 5\}$. Use bootstrap resampling of real returns, with blocks where appropriate, as a robustness check against departures from i.i.d. Gaussian returns. The closed-form optimality and loss results remain specific to the GBM model.
4. **Out-of-sample backtest.** Each month, estimate $\hat{\pi}$ using only the preceding W years of data, hold that fraction for the next month, and roll forward. Compare unconstrained and constrained ($\hat{\pi} \in [0, 1]$) Merton strategies with cash, a 60/40 mix and all-stock using annualised return, volatility, Sharpe ratio, maximum drawdown and realised certainty equivalent, across all four assets and $W \in \{3, 5, 10\}$.

4. Expected outcomes

- A complete derivation of the Merton strategy via the HJB equation, with a closed-form expression for the certainty-equivalent cost of using a wrong portfolio fraction.
- Numerical confirmation that the simulated optimum agrees with π^* within Monte Carlo and discretisation error, and strong convergence orders of about 1/2 (Euler–Maruyama) and 1 (Milstein).
- Evidence that σ is estimated precisely but μ is not, so that $\hat{\pi}$ is highly unstable: with typical data lengths it may range from short selling to leverage.
- A quantified estimation-risk loss, which we expect to be comparable to, or larger than, the entire benefit of optimal investing.
- An assessment of whether the estimated Merton strategy beats simple rules (such as 60/40) on real Indian market data, and whether constraining $\hat{\pi}$ helps.

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